

Session 12 (AIML) - Assignment Questions

Q1. Github Account Setup

1. Create a **GitHub account** (if you don't have one).
 2. Create a new repository named **Python-Basic-Projects**.
 3. After implementing **each feature in each question's solution**, commit your code with a proper commit message (e.g., "Implement Loop for Counting Vowels" or "Create analyze_string function").
 4. After completing the full project, push everything to GitHub.
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Q2. (Strings + Loops + Functions)

Write a function `analyze_string(s)` that takes a string as input and:

- Prints its length using `len()`.
- Prints the string in reverse using slicing.
- Counts and prints how many vowels (a,e,i,o,u) are in the string (case insensitive).
- Uses a for loop with `range()` to print each character with its positive and negative index.

Call this function with user input and handle empty string case.

Q3. (Lists + Functions + Exception Handling)

Create a function `manage_marks()` that:

- Takes 5 subject marks as input from the user.
- Stores them in a list.
- Handles `ValueError` if non-numeric input is given.
- Calculates and prints the average, highest, and lowest marks.
- Sorts the list in descending order and prints it.

Use meaningful comments.

Q4. (OOP + Lists + Exception Handling)

Create a class Student with attributes: name, roll_no, and marks_list (a list of marks).

Include methods:

- `__init__` to initialize the student.
- `add_mark(mark)` to add a mark to the list (handle invalid marks).
- `get_average()` to return the average.
- `display_info()` to show all details.

Create one student object, add marks using user input, and demonstrate all methods with exception handling.

Q5. (Dictionaries + Functions + Control Structures)

Write a function `student_database()` that uses a dictionary (roll number as key) to store student records.

Provide a menu using while loop:

1. Add student (name, age, city)
2. Search student by roll number
3. Display all students
4. Exit

Use `get()`, `update()`, and proper exception handling for invalid inputs.

Q6. (Sets + Tuples + Modules)

Write a program that:

- Takes 10 numbers as input and stores unique numbers in a **set**.
- Converts that set into a **tuple**.
- Uses random module to pick and print 3 random numbers from the tuple.
- Uses math module to print the square root of the sum of the tuple elements.

Handle possible exceptions.

Q7. (Lambda + range() + Lists + Exception Handling)

Create a lambda function to calculate the square of a number.

Write a program that:

- Uses range(1, 21) to generate numbers.
- Stores the squares of these numbers in a list using the lambda function.
- Prints only the even squares from the list.

Add exception handling if needed.

Q8. (Tuples + Dictionaries + OOP)

Create a class Employee with:

- Attributes: emp_id, name, details (a tuple containing department and salary).
- Method show_details() to print all information.

Create a dictionary where employee ID is the key and Employee object is the value.

Add 3 employees and display all using a loop.

Q9. (Strings + Sets + Exception Handling + Modules)

Write a program that:

- Takes a sentence as input.
 - Extracts all unique words (using set) from the sentence.
 - Prints the unique words in sorted order.
 - Uses math module to print the total number of unique words raised to power 2.
 - Handle any possible exceptions gracefully.
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Q10. (Mini Project - Comprehensive)

Create a **Smart Calculator & Data Manager** program that combines multiple concepts:

Use a while loop menu with these options:

1. Basic Arithmetic (use functions + exception handling)
2. Scientific Calculations (use math module)

3. Generate Random Numbers (use random module)
4. Store Results in Dictionary (with timestamp using string)
5. View History (show stored results)
6. Exit